PROJECT OVERVIEW

VEHICLE EFFICIENCY INTELLIGENCE

High-Precision Machine Learning Classification for Global Automotive Performance Benchmarking

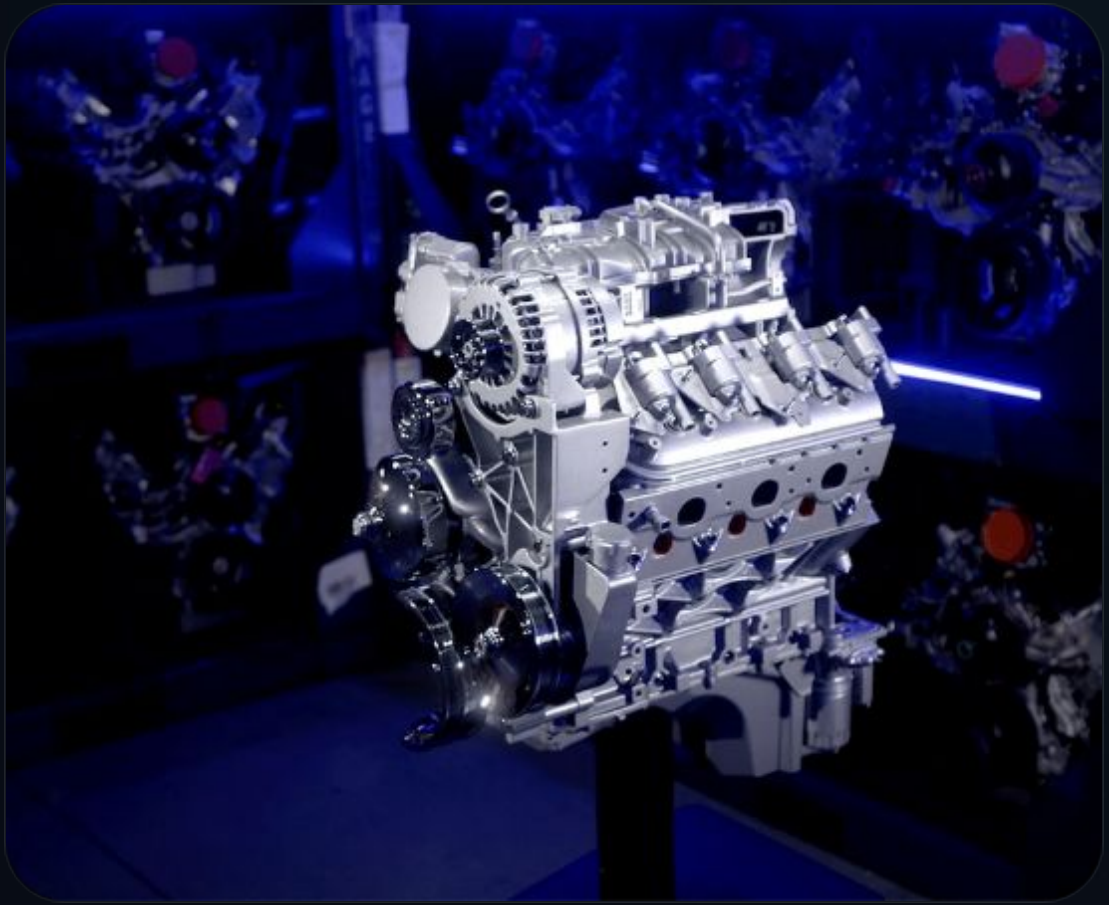
THE DATA ENGINE

Processing **6.5 Million** vehicle records to decode the physics of fuel efficiency.

CORE PERFORMANCE DRIVERS

The model identifies non-linear correlations between four fundamental vehicle metrics:

- **Vehicle Mass:** Measured in kilograms (kg)
- **Engine Capacity:** Volume in CCs
- **Engine Power:** Output in kilowatts (kW)
- **Fuel Systems:** Diesel vs. Petrol logic



CLASSIFICATION STRATEGY

CATEGORY MAPPING

Converted raw KM/L data into operational efficiency classes:

- **High:** > 18 km/l
- **Medium:** 15 - 18 km/l
- **Low:** < 15 km/l

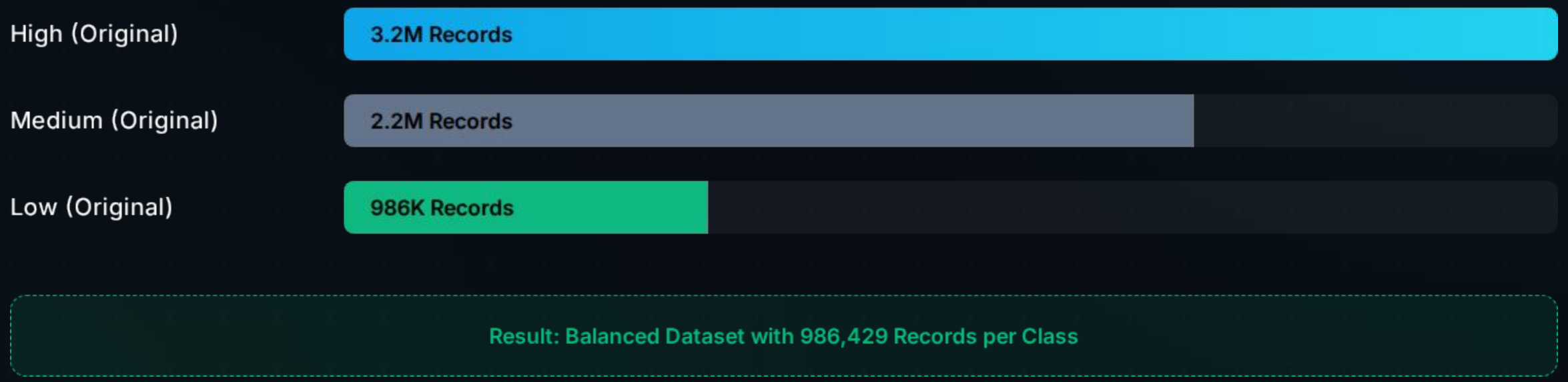
FEATURE ENGINEERING

Optimized input vectors for mathematical consistency:

- **Standard Scaling:** Z-score normalization
- **One-Hot Encoding:** For categorical fuel types
- **Target Encoding:** Label mapping for labels

HANDLING CLASS IMBALANCE

To prevent model bias, we performed **strategic downsampling** to achieve perfect class parity across 3 Million records.



ML DEVELOPMENT LIFECYCLE

CLEANING

Handling nulls & dropping redundant FE columns.

BALANCING

Parity sampling to remove class majority bias.

TRAINING

GridSearch optimization for Random Forest.

EXPORT

Saving .sav model & scalers for deployment.

ALGORITHM BENCHMARKING

Machine Learning Model	Metric Score	F1 (High)	Inference Speed
Random Forest (Winner)	94.4% Accuracy	0.94	Very Fast
Decision Tree	93.9% Accuracy	0.94	Instant
KNN (Manhattan)	93.6% Accuracy	0.94	Slow
Support Vector Machine	79.0% Accuracy	0.81	Very Slow
Logistic Regression	74.0% Accuracy	0.78	Fast

THE ENSEMBLE ADVANTAGE

Why Random Forest surpassed all competitors:

- **Non-Linearity:** Captured complex engine physics.
- **Stability:** Bagging reduced variance and noise.
- **Scale:** Optimized for multi-million record training.

PERFORMANCE METRICS

94.4%

System Accuracy

0.96

Precision Score

TECHNOLOGY STACK

PYTHON CORE

Leveraging Pandas for massive data manipulation and NumPy for vector operations.

SCIKIT-LEARN

Primary engine for algorithm implementation, GridSearch tuning, and metrics.

PERSISTENCE

Serialized deployment using Pickle for model weight storage and versioning.

REAL-WORLD VERIFICATION

PRODUCTION READINESS

The model was validated against popular consumer models with 100% agreement on class labels:

- ✅ Maruti Swift (1.2L): Predicted **HIGH**
- ✅ Honda City (1.5L): Predicted **MEDIUM**
- ✅ Toyota Fortuner (2.8L): Predicted **LOW**

GITHUB LINK

https://github.com/mohanraj-Ai/AI/tree/main/MY%20PROJECTS